

# Test Case 2a: Wing/Body Cruise



- Verification of steady CFD analysis, required
- Settings
  - Steady CFD RANS **French Vanilla SA-[neg] (All terms!)**
    - Adiabatic Wall (not isothermal)
  - Converge residuals to machine precision (~1e-10)
- Grids: [https://dpw.larc.nasa.gov/DPW8/Scatter/Test\\_Case\\_2](https://dpw.larc.nasa.gov/DPW8/Scatter/Test_Case_2)
  - NASA CRM geometry including deformed wing matching condition
    - (L1:Itiny/L2:Coarse/L3:Medium/L4:Fine/L5:eXtra-fine/L6:Ultra-fine)
    - Six-member grid family; four are required, six are desirable
  - Encourage use of committee-supplied grids; user-generated grids are acceptable

## Comparison Data

NTF197: r44,r51,r53  
NTF215: r43,r103  
NTF229: r296,r300,r302  
Ames216: r35,r126,r130,r133

## Reference Units

| Sref (semi-span grid) | Cref     | Semispan   | Moment Center           |
|-----------------------|----------|------------|-------------------------|
| 297360.0 sq.in        | 275.8 in | 1156.75 in | (1325.90, 0.00, 177.95) |

## Conditions

| Mach | Re <sub>c</sub>     | α     | T <sub>static</sub> (120° F) | γ   | Pr   | Pr <sub>t</sub> | Farfield χ = $\tilde{v}/v$ |
|------|---------------------|-------|------------------------------|-----|------|-----------------|----------------------------|
| 0.85 | 5 × 10 <sup>6</sup> | 2.50° | 579.67 R   322.04 K          | 1.4 | 0.72 | 0.90            | 3                          |

## Sutherland's Law

$$\mu(T) = \mu_0 \left(\frac{T}{T_0}\right)^{3/2} \left(\frac{T_0 + S}{T + S}\right) \quad \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{\text{m s}} \quad T_0 = 491.6^\circ \text{R} \quad S = 198.6^\circ \text{R} \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}}\right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}}\right)$$